NUMBER THEORY- COURSEWORK 1

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Read the following instructions carefully:

(1) Solutions should be uploaded to Blackboard by 15:00 on 06/03/2025.
(2) If you are a **3rd year student**, you should hand in solutions to questions 1.1–1.4.
(3) If you are a **4th year student**, work in pairs with another 4th year student and hand in a single document with solutions to questions 1.1–1.5.
(4) To allow the prompt return of scripts and the distribution of feedback and model answers, anyone who is granted an extension of more than 7 calendar days may need to be set an alternative assignment. If this is the case, contact me as soon as possible.

Marking criteria: You will be marked on your ability to present accurate, complete, and concise solutions to the problems set. This includes providing enough working/explanation to demonstrate that appropriate methods have been used, to justify their correctness, and to show that you understand their application. Partial credit is available for questions that show a good understanding of the problem and make a reasonable attempt at giving a solution. Submissions will be marked using the following criteria:

(1) Correctenss of the final answer and for questions with coding elements, how well your code runs (in this case how large a number it can factor): 30% of the marks
(2) Clear and concise explanations of what you are doing and properly justifying why your steps are correct, including referencing of results from lectures (where possible): 70% of marks.
(3) **Marks**: The problem sheet will be marked out of 20. For 3rd years the marks for each question are as follows: 3, 3, 3, 11. For 4th years they are: 2, 2, 2, 4, 10.

Exercise 1.1. Let $K = \mathbb{Q}(\sqrt{7} + \sqrt{-11})$ and let $\alpha = \sqrt{7} + \sqrt{-11}$

(a) Find the minimal polynomial of α.
(b) Find the degree $[K : \mathbb{Q}]$, the norm of α and the trace of α.
(c) How many real/complex conjugate embeddings does the field K have?
(d) Which of the following numbers are algebraic integers? In each case, justify your answer.

$$\frac{\alpha}{3}, \quad \alpha^4 + 10\alpha^3 + 7, \quad \frac{1+\alpha}{2}.$$

(e) Let $F := \mathbb{Q}(\sqrt{7})$ and $E = \mathbb{Q}(\sqrt{-11})$. Find $m_{\alpha,F}$ and $m_{\alpha,E}$.

Exercise 1.2. Find an algebraic number that has integer norm and trace, but it is not an algebraic integer. What is the smallest possible degree of a number field containing any such an element (not necessarily the example you have given)? Justify your answer.

Exercise 1.3. Let K/F be an extension of number fields and assume that $\alpha \in K$ is a root of a monic polynomial with coefficients in $\mathcal{O}_F$. Prove that α is an algebraic integer.

Exercise* 1.4. Let $\alpha = \sqrt[4]{2}$ and let $K = \mathbb{Q}(\alpha)$. Using $\mathrm{Tr}_{K/\mathbb{Q}}$, show that $\sqrt{3}$ is not in K.

Exercise* 1.5. (a) Write some code in Python (or Sage) to factor numbers using the Quadratic Sieve algorithm. Your code should be clearly documented and you should add comments explaining how each part of the algorithm/code works. You may use AI assistants to aid you in writing this code, if so add a brief description at the end exaplainig how you used the AI and any checks you have done to ensure your code is correct.
(b) Using the reading material on blackboard or otherwise, describe how the multiple polynomial quadratic sieve (MPQS) works and modify you code from above to use this improved algorithm.

Super bonus question: **First person to answer this question (before the deadline for coursework 2) will immediately get full marks on both courseworks:**

By any means, factor the following into prime numbers.

642106433294479187895628647812515706243937964211813215270786243050382343053842873793172184604197734118592388512148418164616808434954459903389202584296072475557015329766916149634508596087542261208887882378961